

Alejandro Jaimes

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EDUCATION

University of Central Florida
Bachelor of Science in Computer Science

Orlando, FL
May 2028

EXPERIENCE

President

Apr 2026 - Present

Graphics Programming Knights

Orlando, FL

- Established Graphics Programming Knights, a non-profit student organization, growing membership to **165+** students through programming workshops, learning activities and community socials.
- Organized the first community-wide Render Jam competition resulting in **40+** participants and **5** projects.
- Deliver **two workshops per semester** on pathways into graphics programming, covering topics such as essential mathematics, game development engines, and GPU programming with attendance of **20+** students.
- Expanded club reach by **40%**, establishing partnerships to have workshops and hackathons with **3** organizations.

Outreach Board

Jan 2026 - Present

Knight Hacks

Orlando, FL

- Boosted online community engagement online by **57%** with strategic content creation and event awareness.
- Increased Knight Hacks' activity averaging **5k+** views on Instagram Reels by editing and participating in videos.
- Expanded event attendance by **15%** compared to previous semesters through targeted outreach campaigns on social media and speaking to the public, increasing participation with an average of **40+** people per meeting.

PROJECTS

sFun (SNES emulator) | C++, C, CMake, Assembly, SDL2

- Designed and implemented a **65c816 CPU** emulator supporting, dynamic register width handling using **C++**.
- Built an extensible opcode dispatch system using function pointer tables and implemented core CPU instructions (LDA, REP/SEP, branching, stack operations, JSR/RTS) with correct flag behavior.
- Developed a memory subsystem for SNES RAM/ROM access, enabling instruction fetch, decoding, and execution.
- Engineered robust control flow correctness for relative branching, subroutine calls, and stack-based return handling ensuring **100%** correctness on test cases.

Next Step | React, Tailwind, Next.js, TypeScript, Python, Flask, Vapi

- 3rd place winner** of State Farm's Sponsor Challenge at Shell Hacks 2025 out of **50** projects submitted.
- Created an AI voice agent that delivers interactive insurance quizzes with real-time scoring (0-100) and personalized feedback across **3** insurance categories.
- Implemented backend services to process and structure conversational data generating dynamic agent responses, reducing response time by **66%** using **Python**, integrated with the **Vapi API**.
- Created **7** interactive components for optimal experience with **TypeScript**, **Next.js**, and **React** for frontend.

TK's Revenge | LuaU, Roblox Engine

- 2nd Place winner** at Knight Hacks' Project Launch competition out of **35** projects and **150+** participants.
- Winner of Public Vote** with over **90+** votes, averaging a playthrough of **11** minutes and reached **130+** visits.
- Developed AI Pathfinding scripts in **LuaU** for enemies to attack players, making constant interaction possible.
- Created different movement techniques for **2** different classes to use, ensuring diversified range of options.

Talk to the Hand | C#, Python, Unity, OpenCV, Mediapipe

- Developed in Knight Hacks VIII a 3D shooter game with integrated computer vision hand-tracking system using **OpenCV** and **Mediapipe**, enabling gesture-based gameplay controls.
- Optimized hand gesture recognition improving directional accuracy by **20%** using mathematical equations.
- Programmed a shield mechanic that reduced Game Overs by **30%**, increasing player engagement by **10%**.
- Implemented **4** enemy behaviors and target interactions using **Unity's** physics and scripting APIs.

TECHNICAL SKILLS

Languages: C, C++, C#, Python, HTML, CSS, JavaScript, TypeScript, Assembly, Java

Frameworks/Libraries: Unity, Next.js, Node.js, Tailwind CSS, React, Flask, Unreal Engine, Three.js, OpenCV, Mediapipe, Vapi, SLD2, OpenGL, CUDA, CMake